Conceptual question · Unit 4 · 2 points · 15 min · Conceptual / fast

C04 — Angular momentum under a non-central force and about a moving point

A point particle P of mass m moves with respect to the inertial frame $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with no gravity. $r = |\mathbf{r}_P|$ is its distance to O, $\mathbf{u}_r = \mathbf{r}_P / r$, and $\{\mathbf{e}_R, \mathbf{e}_\varphi, \mathbf{k}\}$ is the cylindrical basis about the z axis. The angular momentum about a point A is $\mathbf{H}_A = \mathbf{r}_{AP} \times m\mathbf{v}_P$ (absolute velocity, Notes Eq. 4.48). Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

A force with an azimuthal part

P moves under $\mathbf{F} = f(r)\,\mathbf{u}_r + q(r)\,\mathbf{e}_\varphi$, with f and q known functions and $q \neq 0$. At $t = 0$, P is in the plane $z = 0$ and its velocity is parallel to that plane. Statement: "Since $\mathbf{F}$ is not central, $\mathbf{H}_O$ is not conserved, and therefore the trajectory of P cannot be planar."

Step 1. The plane $z = 0$ is invariant

In Cartesian components $\mathbf{u}_r = (x, y, z)/r$ and $\mathbf{e}_\varphi = (-y, x, 0)/\sqrt{x^2 + y^2}$, so the z component of Newton's law is

$$m\ddot{z} = f(r)\,\frac{z}{r}$$

With $z(0) = 0$ and $\dot{z}(0) = 0$, the function $z(t) \equiv 0$ satisfies this equation and the initial conditions; by uniqueness it is the solution. P never leaves the plane $z = 0$: the trajectory **is** planar.

Step 2. Angular momentum is not conserved

On the plane, $\mathbf{r}_P = R\,\mathbf{e}_R$ (with $R = r$) and $\mathbf{M}_O = \mathbf{r}_P \times \mathbf{F} = R\,\mathbf{e}_R \times \big(f\,\mathbf{e}_R + q\,\mathbf{e}_\varphi\big) = R\,q(R)\,\mathbf{k} \neq \mathbf{0}$. By Notes Eq. 4.53 ($O$ fixed), $\dot{\mathbf{H}}_O = \mathbf{M}_O$. With $\mathbf{v}_P = \dot{R}\,\mathbf{e}_R + R\dot{\varphi}\,\mathbf{e}_\varphi$:

$$\mathbf{H}_O = R\,\mathbf{e}_R \times m\big(\dot{R}\,\mathbf{e}_R + R\dot{\varphi}\,\mathbf{e}_\varphi\big) = mR^2\dot{\varphi}\,\mathbf{k}, \qquad \frac{d}{dt}\big(mR^2\dot{\varphi}\big) = m\big(2R\dot{R}\dot{\varphi} + R^2\ddot{\varphi}\big) = mR\,a_\varphi = R\,q(R)$$

So $\mathbf{H}_O$ is not conserved (the areal velocity $\tfrac{1}{2}R^2\dot{\varphi}$ is not constant either), but it always points along $\mathbf{k}$.

Step 3. Where the reasoning breaks

Conservation of $\mathbf{H}_O$ is sufficient for planar motion (Notes §4.8.1, property 2) but not necessary. Here $\mathbf{M}_O$ is parallel to $\mathbf{H}_O$: only the magnitude changes, and $\mathbf{r}_P \cdot \mathbf{H}_O = 0$ keeps P in the fixed plane normal to $\mathbf{k}$. The statement is FALSE.

KEY POINTS

- FALSE (the first half is true, the conclusion is not).
- $F_z = f(r)\, z/r$: zero on $z = 0$, so with $z(0) = \dot{z}(0) = 0$ the solution keeps $z \equiv 0$: the motion is planar.
- $\mathbf{M}_O = r\, q(r)\, \mathbf{k} \neq \mathbf{0}$ on the plane: $\mathbf{H}_O = mR^2\dot{\varphi}\, \mathbf{k}$ changes magnitude, $\dfrac{d}{dt}\left(mR^2\dot{\varphi}\right) = R\, q(R)$, but keeps its direction $\mathbf{k}$.

Part (b) · 1 point

Angular momentum about a moving point

Now P moves under a central force towards O on a circle of radius ρ with constant angular rate n: $\mathbf{r}_P = \rho(\cos nt\, \mathbf{i} + \sin nt\, \mathbf{j})$. A point A moves with constant velocity, $\mathbf{r}_A = ut\, \mathbf{i}$ ($u > 0$). Statement: "A frame translating with A at constant velocity is inertial, so the angular momentum of P about A, $\mathbf{H}_A$, is conserved just like $\mathbf{H}_O$."

Step 1. The theorem about a moving point

Differentiating $\mathbf{H}_A = (\mathbf{r}_P - \mathbf{r}_A) \times m\mathbf{v}_P$ with the product rule:

$$\frac{d\mathbf{H}_A}{dt} = (\mathbf{v}_P - \mathbf{v}_A) \times m\mathbf{v}_P + (\mathbf{r}_P - \mathbf{r}_A) \times m\mathbf{a}_P = \mathbf{M}_A - \mathbf{v}_A \times m\mathbf{v}_P$$

because $\mathbf{v}_P \times \mathbf{v}_P = \mathbf{0}$ and $m\mathbf{a}_P = \mathbf{F}$ (Notes Eq. 4.52). Both terms must vanish for conservation. The force passes through O, not through A: $\mathbf{M}_A = (\mathbf{r}_P - \mathbf{r}_A) \times \mathbf{F} = -\mathbf{r}_A \times \mathbf{F}$, since $\mathbf{r}_P \times \mathbf{F} = \mathbf{0}$.

Step 2. Direct computation

$\mathbf{v}_P = n\rho(-\sin nt\, \mathbf{i} + \cos nt\, \mathbf{j})$ and $\mathbf{r}_P - \mathbf{r}_A = (\rho\cos nt - ut)\, \mathbf{i} + \rho \sin nt\, \mathbf{j}$, so

$$H_{A,z} = m\left[(\rho\cos nt - ut)\, n\rho\cos nt + \rho\sin nt\; n\rho\sin nt\right] = m\left(n\rho^2 - ut\, n\rho\cos nt\right)$$

Differentiating with the product rule, $\frac{d}{dt}(t\cos nt) = \cos nt - nt\sin nt$:

$$\dot{H}_{A,z} = -mun\rho\left(\cos nt - nt\sin nt\right), \qquad \dot{H}_{A,z}(0) = -mun\rho \neq 0$$

At $t = 0$, $A = O$ and $\mathbf{M}_A = \mathbf{0}$, yet $\mathbf{H}_A$ changes because of $-\mathbf{v}_A \times m\mathbf{v}_P = -u\,\mathbf{i} \times mn\rho\,\mathbf{j} = -mun\rho\,\mathbf{k}$.

Step 3. The inertial frame of A does not help

In the frame S_A that translates with A (inertial, since $\mathbf{a}_A = \mathbf{0}$), Notes Eq. 4.55 gives $\dot{\mathbf{H}}_{A,A} = \mathbf{M}_A$ for $\mathbf{H}_{A,A} = \mathbf{r}_{AP} \times m(\mathbf{v}_P - \mathbf{v}_A)$. With $\mathbf{F} = -mn^2\mathbf{r}_P$, $\mathbf{M}_A = -ut\,\mathbf{i} \times (-mn^2\rho)(\cos nt\,\mathbf{i} + \sin nt\,\mathbf{j}) = mun^2\rho\,t\sin nt\,\mathbf{k}$, nonzero for instance at $nt = \pi/2$. A frame being inertial does not make the force central about its origin. The statement is FALSE.

KEY POINTS

- FALSE.
- $\mathbf{M}_A = \mathbf{r}_{AP} \times \mathbf{F} = -\mathbf{r}_A \times \mathbf{F} \neq \mathbf{0}$ once $A \neq O$, and $\mathbf{v}_A \times m\mathbf{v}_P \neq \mathbf{0}$.
- $H_{A,z} = m\big(n\rho^2 - ut\,n\rho\cos nt\big)$, $\dot{H}_{A,z} = -mun\rho\,(\cos nt - nt\sin nt)$, equal to $-mun\rho \neq 0$ at $t = 0$.
- Even the relative angular momentum $\mathbf{H}_{A,A}$ of Notes Eq. 4.55 changes: $\dot{H}_{A,A,z} = M_{A,z} = mun^2\rho\,t\sin nt$.

Interpretation and checks

- **Dimensional check:** $mR^2\dot{\varphi}$ and $mn\rho^2$ are kg m^2 s^{-1}; $R\,q(R)$ and $mun\rho$ are N m = kg m^2 s^{-2}.
- **Limits:** in (a), $q \to 0$ gives a central force and $mR^2\dot{\varphi}$ constant; if the initial velocity had a z component, $F_z = fz/r$ would not vanish and the motion would in general leave the plane. In (b), $u \to 0$ (A fixed at O) recovers $\dot{\mathbf{H}}_A = \mathbf{0}$.
- **Common mistakes in the exam:** (1) treating "angular momentum conserved" as the only route to planar motion; (2) applying $\dot{\mathbf{H}}_A = \mathbf{M}_A$ about a moving point without the term $-\mathbf{v}_A \times m\mathbf{v}_P$; (3) thinking that a force is central about every point of an inertial frame.